

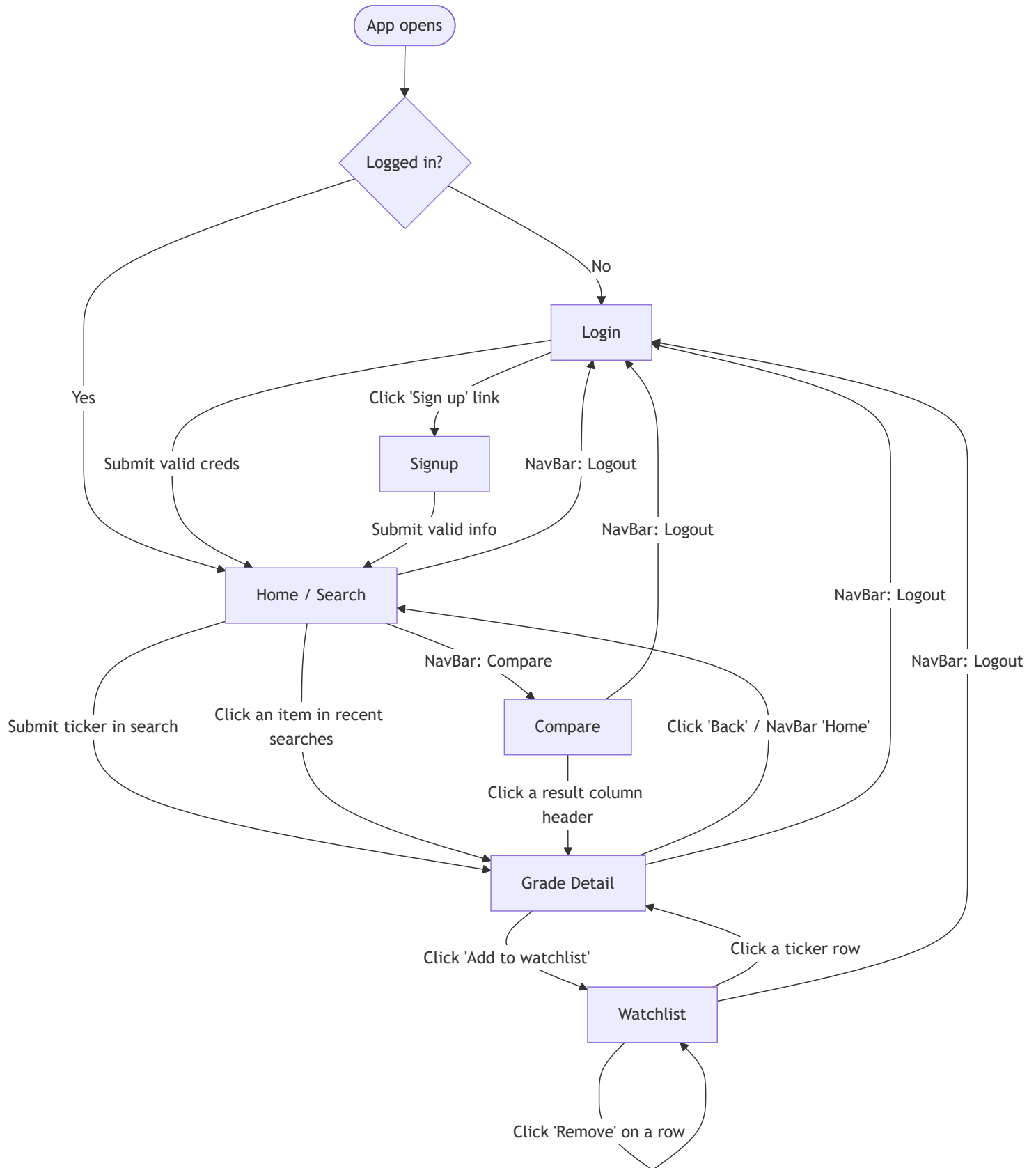
StockGrader — User Flow and Wireframes

Visual version: [wireframes.pdf](#)

Planning artifact for the frontend. Maps every path a user can take through the app, and shows a low-fidelity sketch of every screen.

The wireframes are deliberately rough — boxes, labels, and notes about behaviour. Visual polish comes from Bootstrap during the build phase, not from these sketches.

User Flow



Navigation rules:

- NavBar is visible on every screen.

- When **logged out**, NavBar shows: StockGrader (brand) · Log in · Sign up .
- When **logged in**, NavBar shows: StockGrader · Home · Watchlist · Compare · Welcome, <name> · Logout .
- On phones the NavBar collapses behind a hamburger. The open menu reads top-to-bottom: Welcome, <name> header → Home · Watchlist · Compare → divider → Dark toggle · Logout (responsive flex ordering keeps the desktop bar's left/right layout unchanged).

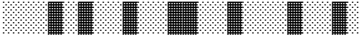
Global page footer (added Class 6 polish):

- Below the main content on every page, a small muted StockGrader © <year> line.
- Year is computed at render time from `new Date().getFullYear()` so it auto-updates each January.

Wireframes

Six screens. For each one: the main element, the actions a user can take, and the four states the instructor requires (happy / loading / empty / error).

1. Login /login

StockGrader [Dark] Log in Sign up <- NavBar	
Log in	What is StockGrader?
[Error alert (red, auto-dismisses after 5s)]	A quick fundamental health check for any publicly traded stock.
[Sign in with Google]	
_____ or _____	Grades A through F based on:
Email	• Topline rev growth
[_____]	• Recent rev (TTM)
Password	• Positive FCF
[_____] [o]	• FCF growth (LT)
^eye	• Recent FCF (TTM)
[Log in]	Score: 5 = A • 0-1 = F
Don't have an account? Sign up	Data from Finnhub, refreshed every 24h.
	<- candlestick band
StockGrader © 2026	

- **Main:** Email + password form on the left, "What is StockGrader?" explainer card on the right. The password field has a **show/hide eye toggle** ([o]) at its trailing edge, styled to match the input.
- **Actions:** submit form, click Google button, click the eye to reveal/hide the password, click "Sign up" link.
- **Layout:** form + card side-by-side at $\geq 768\text{px}$; stack vertically on phones (form first).
- **States:**
 - *happy* → navigate to Home.
 - *loading* → submit button shows "Logging in..." and disabled.
 - *error* → red alert with backend message, **auto-dismisses after 5 seconds**.
 - *empty* → N/A.

2. Signup /signup

```

+-----+
| StockGrader          Log in  Sign up  |
+-----+
| StockGrader          [Dark]  Log in   Sign up  |
+-----+
|
|  Sign up              |  What is StockGrader?  |
|
|  [ Error alert (red, auto- |  A quick fundamental |
|    dismisses after 5s) ]   |  health check for any |
|                             |  publicly traded stock. |
|
|  [ Sign in with Google ]   |
|  _____ or _____ |  Grades A through F   |
|                             |  based on:             |
|  Display name (optional)   |  • Topline rev growth  |
|  [ _____ ]            |  • Recent rev (TTM)    |
|                             |  • Positive FCF        |
|  Email                     |  • FCF growth (LT)     |
|  [ _____ ]            |  • Recent FCF (TTM)    |
|
|  Password (≥ 6 characters) |  Score: 5 = A · 0-1 = F |
|  [ _____ ] [o]        |
|                             ^eye |  Data from Finnhub,   |
|  [ Create account ]        |  refreshed every 24h.  |
|
|  Already registered? Log in |
|
|   <- candlestick band |
|
|  StockGrader © 2026        |
+-----+

```

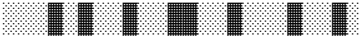
- **Main:** Registration form on the left, "What is StockGrader?" explainer card on the right. The password field has a **show/hide eye toggle** ([o]) at its trailing edge, styled to match the input.
- **Actions:** submit form, click Google button, click the eye to reveal/hide the password, click "Log in" link.
- **Layout:** form + card side-by-side at $\geq 768\text{px}$; stack vertically on phones.
- **States:**
 - *happy* $\rightarrow$ log user in automatically + navigate to Home.

- *loading* → disabled button.
- *error* → red alert (e.g. "email already in use"), **auto-dismisses after 5 seconds**.
- *empty* → N/A.

3. Home /

Home has two layouts depending on whether the user is signed in. Logged-in users get recent searches; logged-out users get the "What is StockGrader?" pitch card beside the search box.

Logged-in (recent searches):

```
+-----+
| StockGrader Home Watchlist Compare [Dark] Logout |
+-----+
|
| Grade a stock
|
| Enter a ticker symbol or company name to get a letter
| grade and a five-point breakdown.
|
| [ e.g. AAPL or Apple ] [ Get grade ]
|
| -----
|
| Recent searches
|
| - MSFT Microsoft Corporation 2026-06-04
| - AAPL Apple Inc. 2026-06-04
| - GOOG Alphabet Inc. 2026-06-01
|
|  <- candlestick band
|
| StockGrader © 2026
+-----+
```

Logged-out (with About card beside search):

+-----+ StockGrader [Dark] Log in Sign up +-----+	
Grade a stock	What is StockGrader?
Enter a ticker symbol or	A quick fundamental
company name to get a letter	health check for any
grade and a five-point	publicly traded stock.
breakdown.	
	Grades A through F
[e.g. AAPL or Apple]	based on:
[Get grade]	• Topline rev growth
	• Recent rev (TTM)
	• Positive FCF
	• FCF growth (LT)
	• Recent FCF (TTM)
	Score: 5 = A · 0-1 = F
	Data from Finnhub,
	refreshed every 24h.
 <- candlestick band	
StockGrader © 2026	
+-----+	

- **Main:** Ticker-or-name search box. Recent searches list when logged in; About card when logged out.
- **Actions:** submit a ticker or company name → navigate to `/grade/:query` . Click a recent search row → navigate to `/grade/:ticker` . Toggle the NavBar dark-mode switch.
- **Layout:** search + About card side-by-side at $\geq 768\text{px}$ when logged out; stack vertically on phones.
- **States:** *happy* → recent searches list (logged in) or About card (logged out) renders. *loading* → "Loading..." placeholder for recent searches. *empty* → "No searches yet — grade your first ticker above." *error* → "Couldn't load recent searches" inline message.

4. Grade Detail `/grade/:query`

The URL accepts either a ticker (`/grade/AAPL`) or a company name (`/grade/Apple`); the backend resolves names to the canonical ticker before fetching.

+-----+		
	StockGrader Home Watchlist Compare [Dark] Logout	
+-----+		
	AAPL \$310.61 USD [Add to watchlist]	
	Apple Inc.	
	+-----+	
	B <- letter grade, big	
	+-----+	
	Criteria	
	[x] Topline revenue growth (long-term)	
	latest: \$416B vs earliest: \$394B	
	[x] Recent revenue growth (TTM)	
	TTM: \$451B vs latest: \$416B	
	[x] Net positive free cash flow	
	latest FCF: \$101B	
	[] Free cash flow growth (long-term)	
	latest: \$99B vs earliest: \$111B	
	[x] Recent free cash flow growth (TTM)	
	TTM: \$112B vs latest: \$101B	
	Last graded: 2026-06-03 08:13 [Refresh]	
+-----+		

- **Main:** Canonical ticker + share price + company name header, letter grade card, 5-criteria checklist with the actual numbers. Two optional banners sit just below the grade card:
 - an **N/A reason** (grey) when the stock can't be graded — e.g. *"Free cash flow can't be computed ... normal for banks, insurers, and other financial firms"* or *"Financial data looks outdated ..."*;
 - a **sector caveat** (amber) on a real grade where free cash flow is only a rough proxy — REITs, insurers, utilities (e.g. *"This is a REIT ... judged on Funds From Operations ..."*).
- **Actions:**
 - "Add to watchlist" → POST /api/watchlist then show inline success/failure alert.
 - "Refresh" → GET /api/grade/<ticker>?refresh=1 bypasses the 24h cache and re-grades, updating the "Last graded" time.
- **States:**

- *happy* → grade + criteria render (with a sector caveat banner for REITs/insurers/utilities).
- *N/A* → letter card shows N/A and the grey reason banner explains why (stale data, or a sector the revenue/FCF model doesn't fit). No criteria list.
- *loading* → spinner with "Grading <ticker> ...".
- *empty* → N/A (a query is always specified by URL).
- *error (logged out)* → friendly info alert: "Please log in to grade stocks" with Log in / Sign up buttons (no scary heading).
- *error (logged in)* → "Couldn't grade ..." red alert.
- *unsupported symbol* → a non-US ticker (e.g. a Toronto .TO listing) shows "StockGrader currently only supports U.S.-listed stocks (NYSE and Nasdaq). Data isn't available for ..." instead of a technical error.

5. Watchlist /watchlist

+-----+								
StockGrader Home Watchlist Compare [Dark] Welcome, Cory Logout								
+-----+								
Your watchlist								
The "Current" grade is the most recent grade we have. Click "Refresh all"								
to update every row with the latest data.								
Add: [e.g. MSFT or Microsoft] [Add]								
[Refresh all] Updated: 8:14:02 AM								
Ticker Name Last price Added At add Now Change								
----- ----- ----- ----- ----- --- -----								
MSFT Microsoft Corp. \$426.28 USD 6/3/2026 B A ▲ Upgraded [Remove]								
AAPL Apple Inc. \$310.61 USD 6/3/2026 B B – No change [Remove]								
TSLA Tesla, Inc. \$228.50 USD 6/1/2026 B D ▼ Downgraded [Remove]								
GOOG Alphabet Inc. – 5/28/2026 – C – (legacy) [Remove]								
+-----+								

◀
▶

- **Main:** Table of saved tickers with company name, last cached price + currency, added date, the grade captured at add time, current cached grade, and an upgrade / downgrade / no-change indicator.
- **Actions:**

- **Add by ticker or name** (POST /api/watchlist) — backend resolves names to canonical tickers and freezes the current grade as `gradeAtAdd` .
- **Remove** a row (DELETE /api/watchlist/:ticker).
- **Refresh all** (POST /api/watchlist/refresh) → re-grade every row at once (cache-bypassed), then show an "Updated <time> " stamp.
- Click the ticker link → navigate to its Grade Detail, where a per-stock **Refresh** button is also available.
- **Responsive behavior:** at phone width (< 576px) the four lower-priority columns (Name, Last price, Added, Grade at add) are hidden — only Ticker, Current, Change, and Remove remain visible so the table fits a 360px viewport without horizontal scroll. All 8 columns return at ≥576px.
- **States:** *happy* → table renders. *loading* → "Loading your watchlist..." *empty* → "Your watchlist is empty. Add a ticker above." *error* → red Bootstrap Alert with the failure reason.

6. Compare /compare

Two view modes once results have loaded — toggle with the **Cards / Table** buttons.

Cards view (default — full breakdown per stock):

```
+-----+
| StockGrader Home Watchlist Compare [Dark] Logout |
+-----+
|
| Compare
| Grade 2 or 3 stocks and see them side by side.
|
| Stock 1 [ AAPL ] Stock 2 [ MSFT ] Stock 3 [ G00G ]
|                               [ Compare ]
|
|                               [ Cards ] [ Table ]
|
| +-----+ +-----+ +-----+
| | AAPL | | MSFT | | G00G |
| | Apple Inc. | | Microsoft Corp. | | Alphabet |
| | $310.61 USD | | $426.28 USD | | $355.50... |
| | B | | B | | B |
| |-----| |-----| |-----|
| | [x] R1 | | [x] R1 | | [x] R1 |
| | [x] R2 | | [x] R2 | | [x] R2 |
| | [x] R3 | | [x] R3 | | [x] R3 |
| | [ ] R4 | | [x] R4 | | [x] R4 |
| | [x] R5 | | [ ] R5 | | [ ] R5 |
| +-----+ +-----+ +-----+
|
+-----+
```

Table view (criteria as rows, stocks as columns — easier to spot where they differ):

+-----+				
[Cards] [Table]				
Criterion	AAPL	MSFT	GOOG	
	Apple...	Micros...	Alpha...	
	\$310.61	\$426.28	\$355.50	
	B	B	B	
-----	-----	-----	-----	
Topline revenue growth (LT)	Yes	Yes	Yes	
Recent revenue growth (TTM)	Yes	Yes	Yes	
Net positive free cash flow	Yes	Yes	Yes	
Free cash flow growth (LT)	No	Yes	Yes	
Recent free cash flow growth	Yes	No	No	
+-----+				

- **Main:** 2–3 tickers (or company names) compared side-by-side. Toggle between **Cards** (one card per stock with full criteria list) and **Table** (criteria as rows, stocks as columns).
- **Actions:**
 - Type tickers/names + submit → fetch via `GET /api/compare?tickers=...` and render.
 - Click **Cards** / **Table** to switch view.
 - Click any stock's column header (Table) or card header (Cards) → navigate to that ticker's Grade Detail.
- **States:** *happy* → toggle + chosen view render. *loading* → spinner with "Comparing stocks...". *empty* → form sits with defaults until submitted. *error* → if one ticker fails, that single card/column shows the error message; the others still render normally.

Consistency rules across all screens

- **NavBar** is identical on every screen and reflects login state.
- **Primary action button** uses the Bootstrap `primary` variant; destructive actions (`Remove` , `Logout`) use a muted style.
- **Error alerts** are a red Bootstrap `Alert` directly above the form or below the page heading.
- **Success alerts** are a green Bootstrap `Alert` directly above the affected element (e.g. "AAPL added to your watchlist.").
- **Banner auto-dismiss:** every `Alert` (success or error) automatically clears itself after 5 seconds via the shared `useAutoDismiss` hook, so banners never linger on screen.

- **Loading states** prefer disabling buttons + showing inline spinners over full-page blockers — feels snappier.
- **Empty states** explain *why* the list is empty and what to do next (never just a blank screen).
- **Mobile-first:** every screen should be usable at ~360px wide. Compare cards stack vertically and the Watchlist hides its lower-priority columns below 576px.
- **Decorative candlestick band** appears below the form on Home (logged out), Login, and Signup — sparse-content pages where it fills the dead space. Data-dense pages (Watchlist, Compare, Grade Detail) skip it.
- **Page footer** (StockGrader © <year> , muted text, year dynamic) sits at the bottom of every page.